

Homework IV

Typed graphs over a fixed type graph form a category

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1 Prerequisites & notation

We work with *directed graphs*. A graph is a quadruple $G = (E, V, s, t)$ where

$$s, t : E \longrightarrow V$$

are the source and target maps (so $s(e)$ is the start vertex of e , and $t(e)$ its end vertex). A *graph morphism* $f : G_1 \rightarrow G_2$ between $G_1 = (E_1, V_1, s_1, t_1)$ and $G_2 = (E_2, V_2, s_2, t_2)$ is a pair of functions $f = (f_E, f_V)$ with $f_E : E_1 \rightarrow E_2$ and $f_V : V_1 \rightarrow V_2$ such that

$$s_2 \circ f_E = f_V \circ s_1 \quad \text{and} \quad t_2 \circ f_E = f_V \circ t_1.$$

Composition in **Graph** is componentwise: $(g \circ f)_E := g_E \circ f_E$ and $(g \circ f)_V := g_V \circ f_V$, and identities are $\text{id}_G = (\text{id}_E, \text{id}_V)$.

Fix once and for all a *type graph*

$$\mathbf{TG} = (E_{\mathbf{TG}}, V_{\mathbf{TG}}, s_{\mathbf{TG}}, t_{\mathbf{TG}}).$$

All typing data in what follows refer to this fixed graph.

2 Typed graphs and their morphisms

Definition 2.1 (Typed graph). A *typed graph over* $\mathbf{TG}$ is a pair (G, type) where $\text{type} : G \rightarrow \mathbf{TG}$ is a graph morphism. Equivalently, $\text{type} = (\text{type}_E, \text{type}_V)$ with

$$s_{\mathbf{TG}} \circ \text{type}_E = \text{type}_V \circ s, \quad t_{\mathbf{TG}} \circ \text{type}_E = \text{type}_V \circ t.$$

Definition 2.2 (Morphisms of typed graphs). Given two typed graphs (G_1, type_1) and (G_2, type_2) , a *typed-graph morphism*

$$f : (G_1, \text{type}_1) \longrightarrow (G_2, \text{type}_2)$$

is a graph morphism $f : G_1 \rightarrow G_2$ such that the *typing triangle* commutes:

$$\text{type}_2 \circ f = \text{type}_1,$$

i.e. componentwise, $\text{type}_{2,E} \circ f_E = \text{type}_{1,E}$ and $\text{type}_{2,V} \circ f_V = \text{type}_{1,V}$.

3 Main result

Definition 3.1 (Category). A *category* $\mathcal{C}$ consists of:

- a class of objects $\text{Ob}(\mathcal{C})$;
- for each pair $A, B \in \text{Ob}(\mathcal{C})$, a *set* $\text{Hom}_{\mathcal{C}}(A, B)$ of morphisms $A \rightarrow B$;

- for each triple A, B, C , a composition operation

$$\circ : \text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C);$$

- for each object A , an identity morphism $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$;

satisfying the axioms:

- **Associativity:** for composable f, g, h we have $(h \circ g) \circ f = h \circ (g \circ f)$;
- **Unit laws:** for $f : A \rightarrow B$, we have $\text{id}_B \circ f = f = f \circ \text{id}_A$.

Theorem 3.2. *The typed graphs over TG with typed-graph morphisms form a category, denoted $\mathbf{Graph}_{/\text{TG}}$.*

Proof. We verify the five requested conditions one by one.

- (1) *Class of objects.* An object is a pair (G, type) where G is a graph and $\text{type} : G \rightarrow \text{TG}$ is a graph morphism. This is well defined: for any fixed G , possible objects with underlying G are in bijection with $\text{Hom}_{\mathbf{Graph}}(G, \text{TG})$, and across all graphs G we obtain a (possibly large) class of objects, which is standard in category theory (large vs. small categories).
- (2) *Set of morphisms between two objects.* Fix two objects (G_1, type_1) and (G_2, type_2) with $G_i = (E_i, V_i, s_i, t_i)$. A graph morphism $f : G_1 \rightarrow G_2$ is a pair of functions $(f_E : E_1 \rightarrow E_2, f_V : V_1 \rightarrow V_2)$ satisfying

$$s_2 \circ f_E = f_V \circ s_1, \quad t_2 \circ f_E = f_V \circ t_1.$$

Such pairs form a set (a subset of the Cartesian product $E_2^{E_1} \times V_2^{V_1}$). The additional *typing* constraints

$$\text{type}_{2,E} \circ f_E = \text{type}_{1,E}, \quad \text{type}_{2,V} \circ f_V = \text{type}_{1,V}$$

pick out a subset of those graph morphisms. Hence

$$\text{Hom}_{\mathbf{Graph}_{/\text{TG}}}((G_1, \text{type}_1), (G_2, \text{type}_2)) = \{ f : G_1 \rightarrow G_2 \text{ in } \mathbf{Graph} \mid \text{type}_2 \circ f = \text{type}_1 \}$$

is a *set*.

- (3) *Composition operation (well-defined and preserves typing).* Let

$$f = (f_E, f_V) : (G_1, \text{type}_1) \rightarrow (G_2, \text{type}_2), \quad g = (g_E, g_V) : (G_2, \text{type}_2) \rightarrow (G_3, \text{type}_3).$$

Define $g \circ f := (g_E \circ f_E, g_V \circ f_V)$.

(a) *$g \circ f$ is a graph morphism.* We check the source equation; the target one is analogous.

As functions $E_1 \rightarrow V_3$,

$$\underbrace{(g_V \circ f_V) \circ s_1}_{\text{LHS}} = g_V \circ (f_V \circ s_1) = g_V \circ (s_2 \circ f_E) = (g_V \circ s_2) \circ f_E = (s_3 \circ g_E) \circ f_E = \underbrace{s_3 \circ (g_E \circ f_E)}_{\text{RHS}},$$

using (in order) associativity of function composition, the fact that f preserves sources, the fact that g preserves sources, and associativity again.

(b) *$g \circ f$ preserves typing.* We must show $\text{type}_3 \circ (g \circ f) = \text{type}_1$ as morphisms $G_1 \rightarrow \text{TG}$, i.e. both on edges and on vertices:

$$\text{type}_{3,E} \circ (g_E \circ f_E) = (\text{type}_{3,E} \circ g_E) \circ f_E = \text{type}_{2,E} \circ f_E = \text{type}_{1,E},$$

since g is typed ($\text{type}_3 \circ g = \text{type}_2$) and f is typed ($\text{type}_2 \circ f = \text{type}_1$). The same argument holds for vertices:

$$\text{type}_{3,V} \circ (g_V \circ f_V) = (\text{type}_{3,V} \circ g_V) \circ f_V = \text{type}_{2,V} \circ f_V = \text{type}_{1,V}.$$

Hence $g \circ f$ is a well-typed morphism $(G_1, \text{type}_1) \rightarrow (G_3, \text{type}_3)$.

- (4) *Identity morphisms (and unit laws).* For any object $(G, \mathbf{type})$ with $G = (E, V, s, t)$, the identity $\text{id}_G = (\text{id}_E, \text{id}_V)$ in **Graph** is a graph morphism because

$$s \circ \text{id}_E = s = \text{id}_V \circ s, \quad t \circ \text{id}_E = t = \text{id}_V \circ t.$$

It is also *typed* since

$$\mathbf{type}_E \circ \text{id}_E = \mathbf{type}_E, \quad \mathbf{type}_V \circ \text{id}_V = \mathbf{type}_V,$$

so $\mathbf{type} \circ \text{id}_G = \mathbf{type}$. Therefore id_G serves as the identity morphism $\text{id}_{(G, \mathbf{type})}$. Finally, for any $f : (G_1, \mathbf{type}_1) \rightarrow (G_2, \mathbf{type}_2)$,

$$\text{id}_{(G_2, \mathbf{type}_2)} \circ f = f = f \circ \text{id}_{(G_1, \mathbf{type}_1)}$$

since composition is inherited componentwise from **Graph**.

- (5) *Associativity of composition.* Given composable typed morphisms f, g, h , we have

$$(h \circ g) \circ f = ((h_E \circ g_E) \circ f_E, (h_V \circ g_V) \circ f_V) = (h_E \circ (g_E \circ f_E), h_V \circ (g_V \circ f_V)) = h \circ (g \circ f),$$

since function composition is associative on both components. Thus composition in **Graph**_{/TG} is associative.

All five conditions hold, so **Graph**_{/TG} is a category. □